Chapter 3

Confidence intervals and hypothesis testing

Exercises

Contents

3 Confidence intervals and hypothesis testing

Exercises **1**
3.1 Concrete items . . . 3
3.2 Aluminum profile . . . 7
3.3 Concrete items (hypothesis testing) . . . 10
3.4 Aluminum profile (hypothesis testing) . . . 13
3.5 Transport times . . . 17
3.6 Cholesterol . . . 24
3.7 Pulse . . . 26
3.8 Foil production . . . 29
3.9 Course project . . . 33
3.10 Concrete items (sample size) . . . 38
3.11 Concrete items 2 . . . 41
3.12 Concrete items 2 (hypothesis testing) . . . 48

3.1 Concrete items

|||| Exercise 3.1 Concrete items

A construction company receives concrete items for a construction. The length of the items are assumed reasonably normally distributed. The following requirements for the length of the elements are made

$$\mu = 3000 \text{ mm}.$$

The company samples 9 items from a delivery which are then measured for control. The following measurements (in mm) are found:

3003	3005	2997	3006	2999	2998	3007	3005	3001

a) Compute the following three statistics: the sample mean, the sample standard deviation and the standard error of the mean, and what are the interpretations of these statistics?

|||| Solution

From the data we get the sample mean and sample standard deviation

$$\bar{x} = 3002.33 \text{ mm} \quad \text{and} \quad s = 3.708 \text{ mm}.$$

From the definition of the standard error of the mean:

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{3.708}{\sqrt{9}} = 1.236.$$

Interpretations:

$\bar{x} = 3002.33$ The best estimate of the true mean length of such concrete items.

$s = 3.708$ The population standard deviation is estimated at 3.7 mm, meaning most items will be within approximately ± 7.4 mm of the mean.

$SE_{\bar{x}} = 1.236$ The estimated standard deviation of the sampling distribution of the sample mean. On average, the sample mean differs from the true population mean by 1.24 mm.

```
x = np.array([3003, 3005, 2997, 3006, 2999, 2998, 3007,
    3005, 3001])
print(x.mean(), x.std(ddof=1), x.std(ddof=1)/np.sqrt(len(x))
    )
```

b) In a construction process, 5 concrete items are joined together to a single construction with a total length equal to the sum of the 5 concrete items. It is very important that the length of this new construction is within 15 m plus/minus 1 cm. How often will it happen that such a construction will be more than 1 cm away from the 15 m target (assume that the population mean concrete item length is $\mu = 3000$ mm and the population standard deviation is $\sigma = 3$)?

|||| Solution

Let $Y = \sum_{i=1}^{5} X_i$ denote the length of the joined construction. Using the rules for linear combinations of independent normals:

$$\begin{aligned} \mathrm{E}(Y) &= 5 \times 3000 = 15000, \\ \mathrm{V}(Y) &= 5 \times 3^2 = 45, \quad Y \sim N(15000,\ 45). \end{aligned}$$

$$P(|Y - 15000| > 10) = 2 \cdot P\left(Z > \frac{10}{\sqrt{45}}\right) = 2 \cdot P(Z > 1.491) = 0.136.$$

About 13–14 out of 100 joined constructions will be more than 1 cm off target.

```
import scipy.stats as stats
print(2*(1 - stats.norm.cdf(10, 0, np.sqrt(45))))
```

c) Find the 95% confidence interval for the mean μ.
(Quantile: $t_{0.975,\,8} = 2.306$*)*

Solution

Using the t-based CI with $\bar{x} = 3002.33$, $s = 3.708$, $n = 9$ and $t_{0.975,8} = 2.306$:

$$3002.33 \pm 2.306 \cdot \frac{3.708}{\sqrt{9}} \iff [2999.5;\ 3005.2].$$

```
ci = stats.t.interval(0.95, df=8, loc=3002.33, scale=3.708/
   np.sqrt(9))
print(ci)
```

d) Find the 99% confidence interval for μ. Compare with the 95% one from above and explain why it is smaller/larger.
(Quantile: $t_{0.995,8} = 3.355$)

Solution

$$3002.33 \pm 3.355 \cdot \frac{3.708}{\sqrt{9}} \iff [2998.2;\ 3006.5].$$

The 99% interval is wider: wanting to be more confident of capturing the true mean μ requires a broader interval.

e) Find the 95% confidence intervals for the variance σ^2 and the standard deviation σ.
(Quantiles: $\chi^2_{0.975,8} = 17.535$, $\chi^2_{0.025,8} = 2.180$)

Solution

Using the χ^2-based formula with $(n-1)s^2 = 8 \times 3.708^2 = 8 \times 13.75 = 110.0$:

$$\left[\frac{8 \times 13.75}{17.535};\ \frac{8 \times 13.75}{2.180}\right] = [6.27;\ 50.46].$$

For the standard deviation:

$$[\sqrt{6.27};\ \sqrt{50.46}] = [2.50;\ 7.10].$$

f) Find the 99% confidence intervals for the variance σ^2 and the standard deviation σ.
(Quantiles: $\chi^2_{0.995,8} = 21.955$, $\chi^2_{0.005,8} = 1.344$*)*

|||| Solution

$$\left[\frac{8 \times 13.75}{21.955};\ \frac{8 \times 13.75}{1.344}\right] = [5.01;\ 81.82].$$

For the standard deviation:

$$[\sqrt{5.01};\ \sqrt{81.82}] = [2.24;\ 9.05].$$

3.2 Aluminum profile

Exercise 3.2 Aluminum profile

The length of an aluminum profile is checked by taking a sample of 16 items whose lengths are measured. The measurement results from this sample are listed below, all measurements are in mm:

180.02	180.00	180.01	179.97	179.92	180.05	179.94	180.10
180.24	180.12	180.13	180.22	179.96	180.10	179.96	180.06

From the data: $\bar{x} = 180.05$ mm and $s = 0.0959$ mm.

It can be assumed that the sample comes from a normally distributed population.

a) Compute the 90%-confidence interval for μ.
Use Python to get the needed quantile from a t-distribution ($t_{1-\alpha/2}$).

Solution

We use Method 3.9.

For the calculation we need $t_{1-\alpha/2}$ from a t-distribution with 15 degrees of freedom (since the number of observations is: $n = 16$, the degrees of freedom are calculated as: $\nu = n - 1 = 15$).

Since we are asked to compute the 90%-confidence interval we need the $t_{0.95}$-quantile (you can convince yourself of this by drawing the pdf of a t-distribution and sketch the interval (centered about the mean) corresponding to 90% of the total probability mass (area under the pdf-curve) - this will leave 5% of the total probability mass on either side of the interval).

We use Python to find the value of the $t_{0.95}$-quantile:

```
print(stats.t.ppf(0.95, df=15))
1.753
```

Using $\bar{x} = 180.05$, $s = 0.0959$, $n = 16$ and $t_{0.95,15} = 1.753$, we now compute the confidence interval:

$$180.05 \pm 1.753 \cdot \frac{0.0959}{\sqrt{16}} = 180.05 \pm 0.042 = [180.01;\ 180.09].$$

b) Compute the 99%-confidence interval for σ.
Use Python to get the needed quantiles from a χ^2-distribution ($\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$).

|||| Solution

We use Method 3.19.

For the calculation we need $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$ from a χ^2-distribution with 15 degrees of freedom ($\nu = n - 1 = 15$).

Since we are asked to compute the 99%-confidence interval we need the $\chi^2_{0.005}$-quantile and the $\chi^2_{0.995}$-quantile (you can convince yourself of this by drawing the pdf of a χ^2-distribution and sketch the interval corresponding to 99% of the total probability mass (area under the pdf-curve), leaving 0.5% of the total probability mass on the left side of the interval and 0.5% of the total probability mass on the right side of the interval).

We use Python to find the values of the quantiles:

```
print(stats.chi2.ppf(0.005, df=15))
4.601

print(stats.chi2.ppf(0.995, df=15))
32.801
```

Using Method 3.19 with $(n-1)s^2 = 15 \times 0.0959^2 = 0.1380$, we get a confidence interval for the variance σ^2:

$$\left[\frac{0.1380}{32.801};\ \frac{0.1380}{4.601}\right] = [0.00421;\ 0.02998].$$

To obtain the confidence interval for the standard deviation, σ, we take the square root:

$$[\sqrt{0.00421};\ \sqrt{0.02998}] = [0.065;\ 0.173].$$

Exam-style questions

c) A sample of $n = 25$ bolts gives the following diameters (in mm):

12.57	12.28	12.67	12.71	12.09
12.23	12.54	12.44	12.50	12.32
12.70	12.68	12.52	12.75	12.61
12.32	12.59	12.30	12.70	12.50
12.47	12.36	12.77	12.47	12.41

From the data: $\bar{x} = 12.50$ mm and $s = 0.18$ mm, assumed normally distributed.
The following computations were performed in Python:

```
print(12.50 - stats.t.ppf(0.975, df=24) * 0.18/np.sqrt
    (25))
12.43

print(12.50 + stats.t.ppf(0.975, df=24) * 0.18/np.sqrt
    (25))
12.57
```

Which of the following is the correct interpretation of these results?

1) 97.5% of all possible sample means from repeated sampling will have values in the interval $[12.43, 12.57]$.
2) 97.5% of individual bolts have diameters in the interval $[12.43, 12.57]$.
3) There is a 95% probability that a newly manufactured bolt's diameter falls in the interval $[12.43, 12.57]$.
4) The true mean μ falls in the interval $[12.43, 12.57]$ with certainty.
5) The interval $[12.43, 12.57]$, is the 95% confidence interval for the population mean μ.

|||| Solution

Correct answer: **5)**.

3.3 Concrete items (hypothesis testing)

|||| Exercise 3.3 Concrete items (hypothesis testing)

This is a continuation of Exercise 1, so the same setting and data is used (read the initial text of it).

a) To investigate whether the requirement to the mean is fulfilled (with $\alpha = 5\%$), test the following hypothesis:

$$H_0 : \mu = 3000$$
$$H_1 : \mu \neq 3000.$$

What is the evidence against the null hypothesis?
(Quantile: $t_{0.975,8} = 2.306$*)*

|||| Solution

The observed t-statistic is:

$$t_{\text{obs}} = \frac{3002.333 - 3000}{3.708/\sqrt{9}} = 1.885.$$

The p-value (using t-distribution with $\nu = 8$ degrees of freedom):

$$p\text{-value} = 2 \cdot P(T > 1.885) = 0.096.$$

There is weak evidence against the null hypothesis. At $\alpha = 0.05$ (critical value ± 2.306) we do not reject H_0; the mean length is consistent with 3000 mm.

```
t_obs = (3002.333 - 3000) / (3.708 / np.sqrt(9))
print(t_obs, 2*(1 - stats.t.cdf(t_obs, df=8)))
```

b) What would the level $\alpha = 0.01$ critical values be for this test, and how are they interpreted?
(Quantile: $t_{0.995,8} = 3.355$*)*

Solution

The critical values are $\pm t_{0.995,8} = \pm 3.355$. For a new experiment, the observed t-statistic must exceed 3.355 in absolute value to lead to a significant result at $\alpha = 0.01$.

c) What would the level $\alpha = 0.05$ critical values be for this test? Compare with the values found in the previous question.
(Quantile: $t_{0.975,8} = 2.306$*)*

Solution

The critical values are $\pm t_{0.975,8} = \pm 2.306$. Compared to the $\alpha = 0.01$ case, it is easier to detect an effect at significance level $\alpha = 0.05$ since the critical value is smaller.

d) Investigate, by means of a normal QQ plot, whether the data appears to come from a normal distribution.

Solution

The nine data points do not deviate from the reference line more than what is typical for truly normally distributed samples of size $n = 9$. We cannot falsify the normality assumption; the data are consistent with a normal distribution.

e) What exactly is plotted in the normal QQ plot? Specifically, what are the x- and y-coordinates of the points?

|||| Solution

The y-coordinates are the ordered observations $x_{(1)} \leq \cdots \leq x_{(9)}$: $2997, 2998, 2999, 3001, 3003, 3005, 3005, 3006, 3007$.

The x-coordinates are theoretical standard normal quantiles. For $n \leq 10$ the formula is $p_i = (i - 3/8)/(n + 1/4)$, giving $z = \Phi^{-1}(p_i)$ for $i = 1, \ldots, 9$.

3.4 Aluminum profile (hypothesis testing)

|||| Exercise 3.4 Aluminum profile (hypothesis testing)

We use the same setting and data as in Exercise 2:

The length of an aluminum profile is checked by taking a sample of 16 items whose lengths are measured. The measurement results from this sample are listed below, all measurements are in mm:

180.02	180.00	180.01	179.97	179.92	180.05	179.94	180.10
180.24	180.12	180.13	180.22	179.96	180.10	179.96	180.06

From the data: $\bar{x} = 180.05$ mm and $s = 0.0959$ mm.

It can be assumed that the sample comes from a normally distributed population.

a) Find the evidence against the following hypothesis (i.e., calculate the p-value and interpret the result):

$$H_0 : \mu = 180.$$

Use Python to compute quantiles and probabilities from relevant probability distribution(s).
Also try using the Scipy function `stats.ttest_1samp(...)` to calculate the p-value directly.

|||| Solution

We use Method 3.23.

First we compute the observed t-statistic:

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{180.05 - 180}{0.0959/\sqrt{16}} = 2.085.$$

t_{obs} represents our *observation* of the stochastic variable T, which is expected to follow a t-distribution with $\nu = n - 1$ degrees of freedom (see also Theorem 3.5).

We now calculate the p-value: the probability of obtaining a value of T that is *at least as extreme* (*at least this far away from zero*) as the test statistic that was actually observed.

The p-value (using t-distribution with $\nu = 15$ degrees of freedom):

$$p\text{-value} = P(T < -2.085) + P(T > 2.085) = 2 \cdot P(T < -2.085)$$

The value is computed using Python:

```
print(2 * stats.t.cdf(-2.085, df = 15))
0.0546
```

So we have p-value $= 0.0546$.

There is "weak evidence against H_0" (see Table 3.1 in the book).

The test statistic and p-value could also be calculated in Python (after having saved the raw data in the variable `x`) in the following way:

```
t_obs, pval = stats.ttest_1samp(x, popmean=180)
print(t_obs, pval)
2.085 0.0546
```

b) If the hypothesis test

$$H_0 : \mu = 180,$$
$$H_1 : \mu \neq 180,$$

is carried out, what are the level $\alpha = 1\%$ critical values?

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

|||| Solution

The critical values (see Definition 3.31) are the 0.005- and 0.995-quantiles of the t-distribution with $\nu = 15$ degrees of freedom.

We use Python to compute the values:

```
print(stats.t.ppf(0.005, df=15))
```

```
-2.94671

print(stats.t.ppf(0.995, df=15))
2.94671
```

This means that if the observed t_{obs} falls within the interval $[-2.94671; 2.94671]$, then we fail to reject the null hypothesis at the 1% significance level (which is the case here since $t_{obs} = 2.085$). This is not surprising since we have already seen that the p-value is indeed larger than 0.01.

c) What is the 99% confidence interval for μ?

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

|||| Solution

To compute the 99% confidence interval we need the quantile $t_{0.995,15} = 2.947$, already found in question b. We use Method 3.9:

$$180.05 \pm 2.947 \cdot \frac{0.0959}{\sqrt{16}} = [179.98;\ 180.12].$$

d) Carry out the hypothesis test

$$H_0 : \mu = 180,$$
$$H_1 : \mu \neq 180,$$

using $\alpha = 5\%$.

|||| Solution

Since we have already found the p-value $= 0.0546$ and concluded that $0.0546 > 0.05 = \alpha$, we fail to reject the null hypothesis at the 5% significance level. In other words, the data do not provide sufficient evidence to conclude that μ is significantly different from 180 mm.

Exam-style questions

e) A textile factory wants to test the thread tensile strength (measured in Newtons) by performing the hypothesis test:

$$H_0 : \mu = 500 \quad \text{vs} \quad H_1 : \mu \neq 500$$

at significance level $\alpha = 5\%$.

The thread tensile strength is then measured in $n = 20$ textile samples.

Without performing the full test, a 95% confidence interval for μ is computed as $[497.2, 503.8]$.

What can be concluded about the hypothesis test?

1) The confidence interval only tells us about μ in general, so we cannot conclude anything about the specific null hypothesis.
2) Since 500 lies inside the interval, the null hypothesis H_0 is rejected at significance level $\alpha = 2.5\%$. We cannot conclude anything about the conclusion at significance level $\alpha = 5\%$.
3) Since 500 lies inside the interval, the null hypothesis H_0 is rejected at significance level $\alpha = 5\%$.
4) Since 500 lies inside the interval, the null hypothesis H_0 cannot be rejected at $\alpha = 5\%$.
5) We would need to know $\bar{x}$ and s separately to determine whether H_0 is rejected.

|||| Solution

Correct answer: **4)**. A two-sided hypothesis test at level α and a $(1 - \alpha)$ confidence interval are two views of the same calculation: $H_0 : \mu = \mu_0$ is rejected at level α if and only if μ_0 falls *outside* the $(1 - \alpha)$ confidence interval for μ. Since $500 \in [497.2, 503.8]$, H_0 is not rejected.

3.5 Transport times

Exercise 3.5 Transport times

A company, MM, selling items online wants to compare the transport times for two transport firms for delivery of the goods. To compare the two companies, recordings of delivery times on a specific route were made, with a sample size of $n = 9$ for each firm. The following data (in days) were found:

Firm A: 2.20, 1.58, 2.41, 2.49, 1.15, 1.46, 2.12, 1.91, 2.05,
Firm B: 0.97, 1.07, 1.02, 1.34, 2.33, 0.88, 2.09, 2.24, 1.47.

It is assumed that data can be regarded as stemming from normal distributions.

The sample average and sample standard deviation has been calculated for each sample:

Firm A: $\bar{y}_A = 1.93$ days, $s_A = 0.45$ days.
Firm B: $\bar{y}_B = 1.49$ days, $s_B = 0.58$ days.

a) We want to test the hypothesis

$$H_0 : \mu_A = \mu_B$$
$$H_1 : \mu_A \neq \mu_B.$$

What is the p-value, interpretation and conclusion for this test (at significance level $\alpha = 5\%$)?

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

Also try using the Scipy function `stats.ttest_ind(...)` to calculate the p-value directly.

Solution

This is the Welch two-sample t-test (we do not assume the data stems from populations with equal variances).

The *estimate* of the difference ($\delta = \mu_A - \mu_B$) is given by $\hat{\delta} = \bar{x}_A - \bar{x}_B$. The standard error of this difference is given by: $\sqrt{s_A^2/n_A + s_B^2/n_B}$. We now consider the (Welch) two sample statistic T:

$$T = \frac{(\bar{X}_A - \bar{X}_B) - \delta_0}{\sqrt{S_A^2/n_A + S_B^2/n_B}}$$

which we know follows (approximately and under the null hypothesis) a t-distribution with ν degrees of freedom, where ν has to be calculated from a somewhat complicated formula given in Theorem 3.50.

We use Method 3.51 to calculate t_{obs} and the corresponding p-value.

The test statistic is:

$$t_{\text{obs}} = \frac{1.93 - 1.49}{\sqrt{0.45^2/9 + 0.58^2/9}} = 1.8$$

The degrees of freedom:

$$\nu = \frac{\left(\frac{0.45^2}{9} + \frac{0.58^2}{9}\right)^2}{\frac{(0.45^2/9)^2}{8} + \frac{(0.58^2/9)^2}{8}} = 15.0$$

The p-value is the probability $P(T < -1.8) + P(T > 1.8)$, given that T follows a t-distribution with $\nu = 15.0$ degrees of freedom (try to sketch the p-value as areas under the pdf-curve in this t-distribution).

Due to symmetry we have:

$$p\text{-value} = P(T < -1.8) + P(T > 1.8) = 2 \cdot P(T < -1.8) = 2 \cdot F(-1.8) = 0.092$$

Where the value of this probability has been computed using Python:

```
print(2 * stats.t.cdf(-1.8, df=15))
0.092
```

Conclusion: There is "weak evidence against the null hypothesis" (see table 3.1).
At $\alpha = 0.05$ we cannot reject H_0; the two firms are not shown to differ significantly.

The test statistic and p-value could also be calculated in Python in the following way:

```
yA = np.array([2.20, 1.58, 2.41, 2.49, 1.15, 1.46, 2.12,
    1.91, 2.05])
```

```
yB = np.array([0.97, 1.07, 1.02, 1.34, 2.33, 0.88, 2.09,
   2.24, 1.47])

t_obs, pval = stats.ttest_ind(yA, yB, equal_var=False)
print(t_obs, pval)

1.798 0.09219
```

Notice that `equal_var = FALSE`, ensures that not assume equal variances in the two polutations (Welch test).

b) Find the 95% confidence interval for the mean difference $\mu_A - \mu_B$.

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

|||| Solution

We use Method 3.47.

For the 95% confidence interval we have $\alpha = 0.05$ and therefore $1 - \alpha/2 = 0.975$. We therefore need to find the 0.975-quantile from a t-distribution with $\nu = 15.0$ degrees of freedom (ν was already calculated in the previous question). We use Python fo find the value of $t_{0.975}$:

```
print(stats.t.ppf(0.975, df=15))
2.131
```

Now we calculate the 95% confidence interval:

$$(1.93 - 1.49) \pm 2.131\sqrt{0.45^2/9 + 0.58^2/9} = [-0.082;\ 0.962].$$

We see that the 95% confidence interval includes the value zero, hence agreeing with our previous conclusion that the two firms are not shown to differ significantly (the difference in means is *not significantly different from zero*, at significance level $\alpha = 0.05$).

c) Use Python to calculate the power of a study with $n = 9$ observations in each of the two samples for detecting a potential mean difference of 0.4 between the firms (assume $\sigma = 0.5$ and $\alpha = 0.05$).
Hint: see example 3.68 in the book.

Solution

We need to use the Python function `smp.TTestIndPower().solve_power(...)` to estimate the power in the scenario of having two independent samples. The function needs the following inputs (leaving out `power`):

`effect_size`: standardized "effect size", i.e., the difference between the two means divided by the standard deviation (in our case this equals $(\mu_0 - \mu_1)/\sigma = 0.4/0.5$)

`nobs1`: The number of observations in sample 1 (in our case this equals 9)

`alpha`: The significance level (in our case this equals 0.05)

Hence we write the following in Python:

```
print(smp.TTestIndPower().solve_power(effect_size = 0.4/0.5,
    nobs1=9, alpha=0.05))
0.35794895
```

So the power is only 0.36 - not nearly good enough for a reasonable study.

d) Use Python to calculate what effect size ($(\mu_0 - \mu_1)/\sigma$) could be detected with $n = 9$ observations in each sample with a power of 0.80 (assume $\sigma = 0.5$ and $\alpha = 0.05$)?
Hint: see example 3.68 in the book.

Solution

Again we need to use the Python function `smp.TTestIndPower().solve_power(...)` to estimate the effect size. This time we give the following inputs (leaving out `effect_size`):

`nobs1`: The number of observations in sample 1 (in our case this equals 9)

`alpha`: The significance level (in our case this equals 0.05)

`power`: The power (in our case this equals 0.80)

Hence we write the following in Python:

```
print(smp.TTestIndPower().solve_power(nobs1=9, alpha=0.05,
    power=0.80))
1.40692
```

If we want to calculate the actual mean difference detectable by the study we compute:

$$(\mu_0 - \mu_1) = \sigma \cdot 1.40692 = 0.5 \cdot 1.40692 = 0.703$$

A mean difference of approximately 0.70 days is detectable with probability 0.80.

e) Use Python to calculate how large a sample size from each firm is needed to detect a potential mean difference of 0.4 with probability 0.90 (assume $\sigma = 0.5$ and $\alpha = 0.05$).
Hint: see example 3.68 in the book.

|||| Solution

Again we need to use the Python function `smp.TTestIndPower().solve_power(...)` to estimate the needed sample sizes. This time we give the following inputs (leaving out `nobs1`):

`effect_size`: standardized "effect size", i.e., the difference between the two means divided by the standard deviation (in our case this equals $(\mu_0 - \mu_1)/\sigma = 0.4/0.5$)

`alpha`: The significance level (in our case this equals 0.05)

`power`: The power (in our case this equals 0.90)

Hence we write the following in Python:

```
print(smp.TTestIndPower().solve_power(effect_size = 0.4/0.5,
    alpha=0.05, power=0.80))
33.8255
```

This is the needed sample size of sample 1. Since we cannot have a non-integer number of observations we need to round up - so we need $n = 34$ observations from each sample (from each firm).

Exam-style questions

f) An electronics manufacturer is evaluating whether a new circuit design reduces power consumption compared with the current design. The engineers plan to compare two independent groups of devices: one group using the existing design (Design A) and one group using the new design (Design B).

From a pilot study, the standard deviation of power consumption is estimated to be $\sigma = 8$ W (watt) for both designs.

The engineers want to be able to detect a difference in mean power consumption of $\mu_A - \mu_B = 5$ W with a significance level of $\alpha = 0.05$ and a power of $1 - \beta = 0.80$.

The study will be conducted using two independent samples, with n devices tested from Design A and n devices tested from Design B.

Which of the following Python codes calculates n correctly and also states the correct conclusion?

1)

```
print(smp.TTestIndPower().solve_power(effect_size =
    0.5, alpha=0.025, power=0.20))
16.96
```

Conclusion: 17 devices must be tested for each circuit design.

2)

```
print(smp.TTestIndPower().solve_power(effect_size =
    0.5, alpha=0.05, power=0.20))
10.96
```

Conclusion: 11 devices must be tested for each circuit design.

3)

```
print(smp.TTestIndPower().solve_power(effect_size =
    0.625, alpha=0.05, power=0.80))
41.169
```

Conclusion: 42 devices must be tested for each circuit design.

4)

```
print(smp.TTestIndPower().solve_power(effect_size =
    0.625, alpha=0.025, power=0.80))
49.94
```

Conclusion: 50 devices must be tested for each circuit design.

5)

```
print(smp.TTestIndPower().solve_power(effect_size =
   0.5, alpha=0.05, power=0.80))
63.77
```

Conclusion: 64 devices must be tested for each circuit design.

|||| Solution

Correct answer: **3)**.
(effect size: $5/8 = 0.625$)

3.6 Cholesterol

Exercise 3.6 Cholesterol

In a clinical trial of a cholesterol-lowering agent, 15 patients' cholesterol (in mmol/L) has been measured before treatment and 3 weeks after starting treatment. Data are listed in the following table:

Patient	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Before	9.1	8.0	7.7	10.0	9.6	7.9	9.0	7.1	8.3	9.6	8.2	9.2	7.3	8.5	9.5
After	8.2	6.4	6.6	8.5	8.0	5.8	7.8	7.2	6.7	9.8	7.1	7.7	6.0	6.6	8.4

It is assumed that data can be regarded as stemming from normal distributions.

The following calculations have been carried out in Python. The researcher that performed the calculations was not sure which Python function to use and decides to compute results using several different t-tests:

```
x1 = np.array([9.1, 8.0, 7.7, 10.0, 9.6, 7.9, 9.0, 7.1, 8.3,
   9.6, 8.2, 9.2, 7.3, 8.5, 9.5])
x2 = np.array([8.2, 6.4, 6.6, 8.5, 8.0, 5.8, 7.8, 7.2, 6.7,
   9.8, 7.1, 7.7, 6.0, 6.6, 8.4])

# 1
t_obs,p_val = stats.ttest_ind(x1,x2,equal_var=False)
print(t_obs, p_val)
3.3206371615181602 0.00257768510995079

# 2
t_obs,p_val = stats.ttest_ind(x1,x2,equal_var=True)
print(t_obs, p_val)
3.3206371615181602 0.00250466153396094

# 3
t_obs,p_val = stats.ttest_rel(x1,x2)
print(t_obs,p_val)
7.340653674886342 3.67225186459351e-06
```

a) Which calculations are relevant and can there, based on these data, be demonstrated a significant decrease in cholesterol levels with $\alpha = 0.001$?

Solution

This is a *paired* setting (same patients measured twice), so only the results from the last part of the Python-calls are relevant (1st calculation assumes a *Welch* two sample setup, 2nd calculation assumes a *pooled* two sample setup, and 3rd calculation assumes a *paired* setup).

A paired t-test gives a p-value of 0.0000037, providing very strong evidence against the null hypothesis. We conclude that the mean cholesterol level has decreased significantly after 3 weeks.

3.7 Pulse

Exercise 3.7 Pulse

13 runners had their pulse measured at the end of a workout and again 1 minute later, with the following results:

Runner	1	2	3	4	5	6	7	8	9	10	11	12	13
Pulse end	173	175	174	183	181	180	170	182	188	178	181	183	185
Pulse 1 min	120	115	122	123	125	140	108	133	134	121	130	126	128

It is assumed that data can be regarded as stemming from normal distributions.

The following was run in Python:

```
Pulse_end = np.array([173, 175, 174, 183, 181, 180, 170, 182,
    188, 178, 181, 183, 185])
Pulse_1min = np.array([120, 115, 122, 123, 125, 140, 108,
   133, 134, 121, 130, 126, 128])
Differences = Pulse_end - Pulse_1min

print(np.mean(Pulse_end))
179.46

print(np.mean(Pulse_1min))
125.0

print(np.mean(Differences))
54.46

print(np.std(Pulse_end,ddof=1))
5.190

print(np.std(Pulse_1min,ddof=1))
8.406

print(np.std(Differences,ddof=1))
5.768

print(stats.t.ppf(0.99, df=12))
2.681

print(stats.t.ppf(0.995, df=12))
3.055
```

a) What is the 99% confidence interval for the mean pulse drop (the drop during 1 minute from end of workout)?

(Try calculating this confidence interval using only the information above and a pocket calculator).

|||| Solution

This is a paired setting: each runner is measured twice, so we work with the differences $D_i = \text{Pulse end}_i - \text{Pulse 1 min}_i$ (the pulse drop during that minute). With $n = 13$, $\bar{x}_D = 54.46$, $s_D = 5.768$ and $t_{0.995,12} = 3.055$, a 99% CI for the mean pulse drop μ_D is

$$\bar{x}_D \pm t_{0.995,12} \cdot \frac{s_D}{\sqrt{n}} = 54.46 \pm 3.055 \cdot \frac{5.768}{\sqrt{13}} = [49.57;\ 59.35].$$

b) Consider the 13 pulse end measurements (first row in the table). What is the 95% confidence interval for the standard deviation of these?

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

|||| Solution

We use Method 3.19.

The 95% CI for σ is based on

$$\left[\sqrt{\frac{(n-1)s^2_{\text{end}}}{\chi^2_{0.975,\,n-1}}};\ \sqrt{\frac{(n-1)s^2_{\text{end}}}{\chi^2_{0.025,\,n-1}}}\right].$$

With $n = 13$, $s_{\text{end}} = 5.190$, so $(n-1)s^2_{\text{end}} = 12 \times 5.190^2 = 323.2$

We need to find the relevant quantiles using Python:

```
print(stats.chi2.ppf(0.025, df=12))
4.404
```

```
print(stats.chi2.ppf(0.975, df=12))
23.34
```

Now we compute the interval:

$$\left[\sqrt{\frac{323.2}{23.34}};\ \sqrt{\frac{323.2}{4.404}}\right] = [3.72;\ 8.57].$$

3.8 Foil production

Exercise 3.8 Foil production

In the production of a certain foil (film), the foil is controlled by measuring the thickness in a number of points distributed over the width. The production is considered stable if the mean of the difference between the maximum and minimum measurements does not exceed 0.35 mm. At a given day, the following random samples are observed for 10 foils:

Foil	1	2	3	4	5	6	7	8	9	10
Max. (mm)	2.62	2.71	2.18	2.25	2.72	2.34	2.63	1.86	2.84	2.93
Min. (mm)	2.14	2.39	1.86	1.92	2.33	2.00	2.25	1.50	2.27	2.37
Max−Min (D)	0.48	0.32	0.32	0.33	0.39	0.34	0.38	0.36	0.57	0.56

It is assumed that data can be regarded as stemming from normal distributions.

The following statistics may potentially be used:

$$\bar{y}_{\max} = 2.508,\ \bar{y}_{\min} = 2.103,\ s_{y_{\max}} = 0.3373,\ s_{y_{\min}} = 0.2834,\ s_D = 0.09664.$$

a) What is a 95% confidence interval for the mean difference μ_D?

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

Also try using the Scipy function
`stats.`*"relevant t-test function"*`.confidence_interval(...)`
to calculate the confidence interval directly.

Solution

The mean difference is $\bar{y}_D = \bar{y}_{\max} - \bar{y}_{\min} = 0.405$, with $s_D = 0.09664$ and $n = 10$.

We use Method 3.9.

For the calculation we need $t_{1-\alpha/2}$ from a t-distribution with 9 degrees of freedom and using $\alpha = 0.05$. Hence we need to find the quantile $t_{0.975,9}$:

```
print(stats.t.ppf(0.975, df=9))
2.262
```

Inserting in the formula we get:

$$\bar{y}_D \pm t_{0.975,9} \cdot \frac{s_D}{\sqrt{n}} = 0.405 \pm 2.262 \cdot \frac{0.09664}{\sqrt{10}} = [0.336;\ 0.474].$$

We can also use Python to calculate the confidence interval directly from raw data:

```
x1 = np.array([2.62, 2.71, 2.18, 2.25, 2.72, 2.34, 2.63,
   1.86, 2.84, 2.93])
x2 = np.array([2.14, 2.39, 1.86, 1.92, 2.33, 2.00, 2.25,
   1.50, 2.27, 2.37])

print(stats.ttest_rel(x1, x2).confidence_interval(0.95))
ConfidenceInterval(low=0.3358, high=0.4741)
```

b) How much evidence is there that the mean difference is different from 0.35? State the null hypothesis, t-statistic and p-value for this question.

|||| Solution

We test $H_0 : \mu_D = 0.35$ against $H_1 : \mu_D \neq 0.35$.

Notice that the 95% CI contains $\mu_D = 0.35$, so we already know the null hypothesis will not be rejected at $\alpha = 0.05$ (which tells us the p-value **must** be larger than 0.05).

The test statistic is

$$t_{\text{obs}} = \frac{\bar{y}_D - 0.35}{s_D/\sqrt{n}} = \frac{0.405 - 0.35}{0.09664/\sqrt{10}} = 1.80,$$

The p-value is the probability $P(T < -1.80) + P(T > 1.80) = 2 \cdot P(T < -1.80)$, given that T follows a t-distribution with $\nu = 9$ degrees of freedom (try to sketch the p-value as areas under the pdf-curve in this t-distribution).

$$p\text{-value} = 2 \cdot P(T < -1.80) = 2 \cdot F(-1.80) = 0.105$$

Where the value of this probability has been computed using Python:

```
print(2 * stats.t.cdf(-1.80, df=9))
0.105
```

Since $p = 0.105 > 0.05$, there is little or no evidence against H_0: we cannot conclude that the mean difference deviates from 0.35. This agrees with the previous question, where the 95% CI, $[0.336; 0.474]$, contained 0.35.

We can also compute the p-value directly using Python:

```
x1 = np.array([2.62, 2.71, 2.18, 2.25, 2.72, 2.34, 2.63,
   1.86, 2.84, 2.93])
x2 = np.array([2.14, 2.39, 1.86, 1.92, 2.33, 2.00, 2.25,
   1.50, 2.27, 2.37])
diff_data = x1 - x2

t_obs, p_val = stats.ttest_1samp(diff_data, popmean=0.35)
print(t_obs, p_val)
1.79976 0.10543
```

Notice that we use the one sample test (`ttest_1samp`) instead of the inbuilt function for a paired setup (`ttest_rel`), because this lets us define a null hypotheis for the difference that is not exactly zero (`popmean=0.35`).

Exam-style questions

c) Continuing on the exercise above it has been estimated that the standard deviation of the thickness differences is $\sigma_D \approx 0.097$ mm. The sample ($n = 10$) lead to a 95% confidence interval with a margin of error of approximately ≈ 0.07 mm. A quality engineer wants to carry out a new study to estimate the mean difference with a margin of error no larger than 0.03 mm.
What is the minimum sample size needed?

1) $n = 39$
2) $n = 10$
3) $n = 41$
4) $n = 24$
5) $n = 40$

Solution

We use Method 3.63.

$$n = \left(\frac{z_{1-\alpha/2} \cdot \sigma}{ME}\right)^2 = \left(\frac{1.96 \times 0.097}{0.03}\right)^2 = 40.16,$$

We must round up to 41.

Correct answer: **3)**.

3.9 Course project

|||| Exercise 3.9 Course project

At a specific education it was decided to introduce a project, running through the course period, as part of the grade point evaluation. In order to assess whether it has changed the percentage of students passing the course, the following data was collected:

	Before introduction of project	After introduction of project
Number of students evaluated	50	24
Number of students failed	13	3
Average grade point $\bar{x}$	6.420	7.375
Sample standard deviation s	2.205	1.813

a) Assuming that the grades are approximately normally distributed in each group, test the hypothesis:

$$H_0 : \mu_{\text{Before}} = \mu_{\text{After}},$$
$$H_1 : \mu_{\text{Before}} \neq \mu_{\text{After}}.$$

State the test statistic, p-value and conclusion (at $\alpha = 0.05$).

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

|||| Solution

Since the two groups have quite different sample sizes and standard deviations, we use Welch's t-test:

$$t_{\text{obs}} = \frac{\bar{x}_{\text{Before}} - \bar{x}_{\text{After}}}{\sqrt{s^2_{\text{Before}}/n_{\text{Before}} + s^2_{\text{After}}/n_{\text{After}}}} = \frac{6.420 - 7.375}{\sqrt{2.205^2/50 + 1.813^2/24}} = -1.973$$

The degrees of freedom are also found from the following equation:

$$\nu = \frac{\left(\frac{s^2_{\text{Before}}}{n_{\text{Before}}} + \frac{s^2_{\text{After}}}{n_{\text{After}}}\right)^2}{\frac{\left(s^2_{\text{Before}}/n_{\text{Before}}\right)^2}{n_{\text{Before}} - 1} + \frac{\left(s^2_{\text{After}}/n_{\text{After}}\right)^2}{n_{\text{After}} - 1}} = \frac{(0.09724 + 0.13696)^2}{0.09724^2/49 + 0.13696^2/23} = \frac{0.05485}{0.001009} \approx 54.385$$

This gives a p-value:

$$p\text{-value} = 2 \cdot P(T > 1.97) = 0.054.$$

Where the value is computed using Python:

```
print(2 * stats.t.cdf(-1.973, df = 54.385))
0.0536
```

At $\alpha = 5\%$ we cannot reject H_0: there is no significant difference in mean grade point before and after the introduction of the project.

b) Calculate a 99% confidence interval for the mean grade point difference.

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

|||| Solution

For a 99% confidence interval we have $\alpha = 0.01$ and $1 - \alpha/2 = 0.995$.

For the calculation we need the quantile $t_{0.995,54.4}$:

```
print(stats.t.ppf(0.995, df = 54.4))
2.669
```

The 99% CI for $\mu_{\text{Before}} - \mu_{\text{After}}$ is

$$(\bar{x}_{\text{Before}} - \bar{x}_{\text{After}}) \pm t_{0.995,54.4} \sqrt{\frac{s^2_{\text{Before}}}{n_{\text{Before}}} + \frac{s^2_{\text{After}}}{n_{\text{After}}}}.$$

With $\bar{x}_{\text{Before}} - \bar{x}_{\text{After}} = 6.420 - 7.375 = -0.955$ and $t_{0.995,54.4} = 2.669$:

$$-0.955 \pm 2.669\sqrt{2.205^2/50 + 1.813^2/24} = [-2.247;\ 0.337].$$

Note that, as expected, the interval contains 0, consistent with the observation p-value > 0.01, from the previous question.

c) Calculate a 95% confidence interval for the grade point standard deviation *after* the introduction of the project.

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

Solution

Notice that we are only asked to consider the grade point standard deviation *after* the introduction of the project - hence this exercise regards only one sample.

We use Method 3.19.

For the calculation we need $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$ from a χ^2-distribution with $n-1 = 24-1 = 23$ degrees of freedom.

Since we are asked to compute the 95%-confidence interval we need the $\chi^2_{0.025}$-quantile and the $\chi^2_{0.975}$-quantile, leaving 2.5% of the total probability mass on the left side of the interval and 2.5% of the total probability mass on the right side of the interval).

We use Python to find the values of the quantiles:

```
print(stats.chi2.ppf(0.025, df=23))
11.689

print(stats.chi2.ppf(0.975, df=23))
38.076
```

Using $(n-1)s^2 = 23 \times 1.813^2 = 23 \times 3.287 = 75.6$:

$$\left[\sqrt{\frac{(n-1)s^2_{\text{After}}}{\chi^2_{0.975,n-1}}};\ \sqrt{\frac{(n-1)s^2_{\text{After}}}{\chi^2_{0.025,n-1}}}\right] = \left[\frac{\sqrt{23\times 1.813^2}}{\sqrt{38.076}};\ \frac{\sqrt{23\times 1.813^2}}{\sqrt{11.689}}\right] = [1.41;\ 2.54].$$

Exam-style questions

Continuing on the exercise above the teacher would like to collect more data to test whether there may be a small grade improvement after introducing the project. The teacher decides he would like to be able to detect a grade improvement of 0.5 with statistical power of 80% ($1-\beta = 0.80$), when using significance level $\alpha = 0.05$.

Looking into older records the teacher concludes that the usual average grade before introducing the project can be assumed to be 6.4. Since extensive historical data are available from earlier years, the teacher decides to treat the historical mean grade of 6.4 as known and plans a one-sample study of grades after introduction of project.

To avoid underestimating the required sample size, the teacher assumes that the standard deviation of grades after introduction of project will be $\sigma_A = 2.54$.

To solve this exercise you may use one or more of the following quantiles from a standard normal distribution:

$$z_{0.80} = 0.842 \qquad z_{0.85} = 1.036 \qquad z_{0.95} = 1.645 \qquad z_{0.975} = 1.960$$

d) Why does the teacher use the value $\sigma_A = 2.54$ when planning the study?

1) Because 2.54 is the sample standard deviation observed after introducing the project.
2) Because 2.54 is the upper endpoint of the previously calculated 95% confidence interval for the standard deviation after introducing the project, giving a conservative estimate of variability.
3) Because 2.54 is approximately equal to the standard deviation before introducing the project.
4) the value 2.54 is a guess that the teacher must base on information not stated in this exercise.
5) Because 2.54 is the estimated difference in mean grades before and after introducing the project.

|||| Solution

The parameter σ is unknown. In the previous question a 95% confidence interval for the standard deviation after introducing the project was found to be $[1.41; 2.54]$. When planning a new study it is common to use a conservative estimate of the standard deviation. Therefore the teacher uses the upper confidence limit $\sigma_A = 2.54$, ensuring that the required sample size is not underestimated.

e) How many students should the teacher expect to include in the sample in order to achieve a statistical power of 80%?

1) 101
2) 148
3) 176
4) 203
5) 254

|||| Solution

We use Method 3.65:

$$n = \left(\sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu - \mu_0}\right)^2$$

For planning a one-sample test of

$$H_0 : \mu = 6.4, \qquad H_1 : \mu \neq 6.4,$$

with desired detectable difference $\mu - \mu_0 = 0.5$, power $1 - \beta = 0.80$ and significance level $\alpha = 0.05$.

From the information provided in the exercise we must use the values:

$$z_{1-\beta} = 0.842 \qquad \text{and} \qquad z_{1-\alpha/2} = 1.960$$

Using $\sigma = 2.54$:

$$n = \left(\sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu - \mu_0}\right)^2 = \left(2.54 \frac{0.842 + 1.960}{0.5}\right)^2$$

This gives

$$n = (2.54 \times 5.604)^2 = 202.61.$$

which we must round up tot 203.

Correct answer: **4)**

3.10 Concrete items (sample size)

|||| Exercise 3.10 Concrete items (sample size)

This is a continuation of Exercise 1, so the same setting and data is used (read the initial text of it).

a) A study is planned for a new supplier. It is expected that the standard deviation will be approximately $\sigma = 3$ mm. We want a 90% confidence interval for the mean value in this new study to have a width of 2 mm. How many items should be sampled to achieve this?
(Quantile: $z_{0.95} = 1.645$*)*

|||| Solution

Using the sample size formula with wanted margin of error $ME = 1$ (half the width):

$$n = \left(\frac{z_{0.95} \cdot \sigma}{ME}\right)^2 = \left(\frac{1.645 \times 3}{1}\right)^2 = 24.35.$$

Hence at least $n = 25$ items must be sampled.

b) Answer the sample size question above but requiring the 99% confidence interval to have the same width of 2 mm.
(Quantile: $z_{0.995} = 2.576$*)*

|||| Solution

$$n = \left(\frac{z_{0.995} \times 3}{1}\right)^2 = \left(\frac{2.576 \times 3}{1}\right)^2 = 59.72.$$

Hence at least $n = 60$ items must be sampled.

c) (Challenging) For the two sample sizes found above, find the probability that the corresponding confidence interval in a future study will be more than 10% wider than planned (still assuming $\sigma^2 = 9$).

Solution

The random half-width of the CI is $t_{1-\alpha/2} \cdot S/\sqrt{n}$. Since $(n-1)S^2/9 \sim \chi^2_{n-1}$, the probability that the half-width exceeds $1.1 \times ME$ can be written as a tail probability of the χ^2 distribution.

For $\alpha = 0.10$ and $n = 25$ (so $t_{0.95,24} = 1.711$):

$$P\left(t_{0.95} \cdot \frac{S}{\sqrt{25}} > 1.1\right) = P\left(\chi^2_{24} > \frac{1.1^2 \times 24 \times 25}{1.711^2 \times 9}\right) = P(\chi^2_{24} > 27.56) = 0.28.$$

Almost 30% of the time the planned experiment will yield a CI wider than intended.

For $\alpha = 0.01$ and $n = 60$ (so $t_{0.995,59} = 2.662$):

$$P(\chi^2_{59} > 68.0) \approx 0.22.$$

In this case it only happens in about 22% of cases.

d) With the sample size $n = 25$ from the first question, what are the probabilities of detecting (i.e. rejecting $H_0 : \mu = 3000$) when the true mean is $\mu_1 = 3001, 3002$, or 3003, respectively? Assume $\alpha = 0.05$ and $\sigma = 3$.

Solution

Using the power of a one-sample t-test:

$$\delta = |\mu_1 - \mu_0| \in \{1, 2, 3\}.$$

```
from statsmodels.stats.power import TTestPower
for d in [1, 2, 3]:
    print(d, TTestPower().power(d/3, nobs=25, alpha=0.05))
```

The three powers are approximately 0.36, 0.89 and 0.998. A difference of 1 mm is unlikely to be detected, but differences of 2 and 3 mm have high detection probability.

e) Answer the same question as above but using $n = 60$.

Solution

```
for d in [1, 2, 3]:
    print(d, TTestPower().power(d/3, nobs=60, alpha=0.05))
```

The three powers are approximately 0.72, 0.999 and 1.00. With $n = 60$ even a difference of 1 mm has a reasonable detection probability.

f) What sample size is needed to achieve a power of 0.80 for detecting an effect of size 0.5 mm (assume $\sigma = 3$ and $\alpha = 0.05$)?

Solution

```
n = TTestPower().solve_power(effect_size=0.5/3, power=0.8,
    alpha=0.05)
print(np.ceil(n))
```

Using the formula: $n = \left(3 \cdot \frac{z_{0.8}+z_{0.975}}{0.5}\right)^2 \approx 283$, or more precisely $n = 285$ from the t-test power function. A difference of 0.5 mm is expensive to detect reliably.

g) Assume you only have finances for $n = 50$ observations. How large a difference would you be able to detect with probability 0.80?

Solution

```
delta = TTestPower().solve_power(nobs=50, power=0.8, alpha
    =0.05)*3
print(delta)
```

The detectable effect is approximately 1.21 mm. A true mean of 3001.2 or higher (or 2998.8 or lower) would be detected with probability 0.80.

3.11 Concrete items 2

|||| Exercise 3.11 Concrete items 2

A construction company receives concrete items for a construction. The lengths of the items are assumed reasonably normally distributed, with mean $\mu = 3000$ mm and standard deviation $\sigma = 3$ mm.

a) In a construction process, 5 concrete items are joined together to a single construction with a total length equal to the sum of the 5 concrete items. It is very important that the length of this joined construction is within 15 m plus/minus 2 cm. How often will it happen that such a construction will be more than 2 cm away from the 15 m target?

(*Use Python to compute probabilities from a relevant normal distribution*)

|||| Solution

The lengths of the joined constructions are subject to some randomness given they are assembled from 5 individual pieces with random lengths (which follow a normal distribution as described). The total joined length is a sum of 5 individual (but random) lengths. Therefore the length of the joined construction is given by $Y = \sum_{i=1}^{5} X_i$, where X_i are random variables describing the lengths of the individual pieces.

Using the rules for linear combinations of independent normals, we compute the mean and variance of Y:

$$E[Y] = \sum_{i=1}^{5} E[X_i] = 3000 + 3000 + 3000 + 3000 + 3000 = 15000,$$

$$V[Y] = \sum_{i=1}^{5} 1^2 \cdot V[X_i] = 3^2 + 3^2 + 3^2 + 3^2 + 3^2 = 45$$

Since Y is a linear combination of normally distributed stochastic variables, Y will also follow a normal distributions. Hence we have: $Y \sim N(15000,\ 45)$.

Now we want to calculate the probability that Y deviates 2 centimeters (20 mm) or more from the target of 15 meters (15000 mm), i.e., $P(Y > 15020) + P(Y < 14980)$. Since we know that Y follows a normal distribution we have:

$$P(Y > 15020) = 1 - F(15020)$$
$$P(Y < 14980) = F(14980)$$

where $F(Y)$ is the cdf of Y. We use Python to compute these probabilities:

```
print(1 - stats.norm.cdf(15020, 15000, np.sqrt(45)))
0.0014345

print(stats.norm.cdf(14980, 15000, np.sqrt(45)))
0.0014345
```

So the total probability is $0.0014345 + 0.0014345 = 0.00287$. To conclude: the joined construction will be more than 2 cm away from the 15 m target in roughly 0.3% of cases.

The construction company wants to check that the individual concrete items do in fact have an average length of 3000 mm and a standard deviation of 3 mm (still assuming that the lengths follow a normal distribution). The company samples 9 items from a delivery which are then measured for control. The following measurements (in mm) are found:

3003	3005	2997	3006	2999	2998	3007	3005	3001

b) Compute the following three statistics: the sample mean, the sample standard deviation and the standard error of the mean (*you may use Python for calculations*).
Explain the difference between the sample standard deviation and the standard error of the mean.

|||| Solution

```
x = np.array([3003, 3005, 2997, 3006, 2999, 2998, 3007,
   3005, 3001])
print(x.mean(), x.std(ddof=1), x.std(ddof=1)/np.sqrt(9))
3002.33 3.708 1.236
```

From the data we get the sample mean and sample standard deviation

$$\bar{x} = 3002.33 \text{ mm} \quad \text{and} \quad s = 3.708 \text{ mm},$$

and the standard error of the mean

$$SE_{\bar{x}} = \frac{3.708}{\sqrt{9}} = 1.236.$$

The **sample standard deviation** describes the variation between individual concrete items. The **standard error of the mean** instead describes the expected variation of the sample mean ($\bar{x}$) when calculated from samples of the same size ($n = 9$).

c) Find the 95% confidence interval for the mean μ.

(*Use Python to get the needed quantile(s) from a relevant probability distribution*)

|||| Solution

We use Method 3.9.

For the calculation we need $t_{1-\alpha/2}$ from a t-distribution with 8 degrees of freedom ($\nu = n - 1 = 8$).

Since we are asked to compute the 95%-confidence interval we need the $t_{0.975}$-quantile (you can convince yourself of this by drawing the pdf of a t-distribution and sketch the interval (centered about the mean) corresponding to 95% of the total probability mass (area under the pdf-curve) - this will leave 2.5% of the total probability mass on either side of the interval).

We use Python to find the value of the $t_{0.975}$-quantile:

```
print(stats.t.ppf(0.975, df=8))
2.306
```

We can also write: $t_{0.975,\,\nu=8} = 2.306$ (to make it more clear which distribution is used).

The 95% confidence interval for the mean is given as:

$$3002.33 \pm 2.306 \cdot \frac{3.708}{\sqrt{9}} \iff [2999.5;\ 3005.2].$$

d) Find the 99% confidence interval for μ. Compare with the 95% one from above and explain why it is smaller/larger.

(*Use Python to get the needed quantile(s) from a relevant probability distribution*)

Solution

The needed quantile is $t_{0.995,\,\nu=8}$, and we find the value using Python:

```
print(stats.t.ppf(0.995, df=8))
3.355
```

The 99% confidence interval for the mean is given as:

$$3002.33 \pm 3.355 \cdot \frac{3.708}{\sqrt{9}} \iff [2998.2;\ 3006.5].$$

The 99% interval is wider: wanting to be *more confident* of capturing the true mean μ requires a *broader interval*.

e) Find the 95% confidence intervals for the variance σ^2 and the standard deviation σ.

(*Use Python to get the needed quantile(s) from a relevant probability distribution*)

Solution

We use Method 3.19.

For the calculation we need $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$ from a χ^2-distribution with 8 degrees of freedom ($\nu = n - 1 = 8$).

Since we are asked to compute the 95%-confidence interval we need the $\chi^2_{0.025}$-quantile and the $\chi^2_{0.975}$-quantile (you can convince yourself of this by drawing the pdf of a χ^2-distribution and sketch the interval corresponding to 95% of the total probability mass (area under the pdf-curve), leaving 2.5% of the total probability mass on the left side of the interval and 2.5% of the total probability mass on the right side of the interval).

We use Python to find the values of the quantiles: $\chi^2_{0.025,\,\nu=8}$ and $\chi^2_{0.975,\,\nu=8}$:

```
print(stats.chi2.ppf(0.025, df=8))
2.180
```

```
print(stats.chi2.ppf(0.975, df=8))
17.535
```

Using $(n-1)s^2 = 8 \times 3.708^2 = 110.0$:

$$\left[\frac{8 \times 13.75}{17.535};\ \frac{8 \times 13.75}{2.180}\right] = [6.27;\ 50.46].$$

For the standard deviation:

$$[\sqrt{6.27};\ \sqrt{50.46}] = [2.50;\ 7.10].$$

f) Based on the 95% confidence intervals, what should the construction company conclude about the delivered concrete items? Can they trust that the lengths of the individual items have mean 3000 mm and standard deviation 3 mm?

|||| Solution

The company cannot be absolutely sure about the exact true values of μ and σ, but since the 95% CI for μ includes 3000 and the 95% CI for σ includes 3, these values are entirely consistent with the data, even though the sample estimates $\bar{x}$ and s are not exactly equal to 3000 and 3 respectively.

g) Suppose instead the construction company samples $n = 100$ concrete items (instead of 9), and obtains sample estimates similar to before: $\bar{x} = 3002.3$ mm and $s = 3.7$ mm. Based on 95% confidence intervals, what conclusion would the company now reach?

(*Use Python to get the needed quantile(s) from a relevant probability distribution*)

Solution

In this situation the number of observations in the sample is *larger* ($n = 100$), which makes the estimates *more reliable* - this means we would expect *narrower confidence intervals*!

When performing the calculations, the only difference is the degrees of freedom used in the probability distributions: $\nu = 100 - 1 = 99$ (for both the relevant t-distribution and the relevant χ^2-distribution).

For the 95% confidence interval for the mean we need the quantile, $t_{0.975,\nu=99} = 1.984$:

```
print(stats.t.ppf(0.975, df=99))
1.984
```

(*notice the value* $t_{0.975,\nu=99} = 1.984$ *is very close to the corresponding value from a standard normal distribution* $z_{0.975} = 1.96$*, because the degrees of freedom,* ν*, is quite large in this case*).

We now compute the interval:

$$3002.3 \pm 1.984 \cdot \frac{3.7}{\sqrt{100}} \iff [3001.57;\ 3003.03].$$

For the 95% confidence interval for the variance (and standard deviation) we need the quantiles, $\chi^2_{0.025,\nu=99}$ and $\chi^2_{0.025,\nu=99}$:

```
print(stats.chi2.ppf(0.025, df=99))
73.36

print(stats.chi2.ppf(0.975, df=99))
128.42
```

Using $(n-1)s^2 = 99 \times 3.7^2 = 1355.31$:

$$\left[\frac{99 \times 3.7^2}{128.42};\ \frac{99 \times 3.7^2}{73.36}\right] = [10.55;\ 18.47].$$

For the standard deviation:

$$[\sqrt{10.55};\ \sqrt{18.47}] = [3.25;\ 4.30].$$

This gives $\mu \in [3001.6; 3003.0]$ and $\sigma \in [3.25; 4.30]$. Neither interval contains the stated values 3000 mm and 3 mm, even though the point estimates ($\bar{x} = 3002.3$, $s = 3.7$) are almost identical to the $n = 9$ sample above, so the company would now suspect (and likely conclude) that the true population mean and standard deviation

are both somewhat larger than expected. The extra precision from the larger sample reveals a deviation that the small sample above could not.

3.12 Concrete items 2 (hypothesis testing)

|||| Exercise 3.12 Concrete items 2 (hypothesis testing)

This is a continuation of Exercise 11:

A construction company receives concrete items for a construction. The lengths of the items are assumed reasonably normally distributed, with mean $\mu = 3000$ mm and standard deviation $\sigma = 3$ mm.

The company samples 9 items from a delivery which are then measured for control. The following measurements (in mm) are found:

3003	3005	2997	3006	2999	2998	3007	3005	3001

a) To investigate whether the requirement on the mean is fulfilled (with $\alpha = 5\%$), test the following hypothesis:

$$H_0 : \mu = 3000$$
$$H_1 : \mu \neq 3000.$$

Or similarly asked: what is the evidence against the null hypothesis?

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

|||| Solution

From Exercise 11, we already know $\bar{x} = 3002.33$ and $s = 3.708$:

```
x = np.array([3003, 3005, 2997, 3006, 2999, 2998, 3007,
   3005, 3001])
print(x.mean(), x.std(ddof=1))
3002.33 3.708
```

We compute t_{obs} (Method 3.23):

$$t_{\text{obs}} = \frac{3002.33 - 3000}{3.708/\sqrt{9}} = 1.885.$$

The p-value (using a t-distribution with $\nu = n - 1 = 8$ degrees of freedom) is then calculated as (Method 3.23):

$$p\text{-value} = 2 \cdot P(T > 1.885) = 2 \cdot P(T < -1.885) = 0.096$$

where the actual value was computed using Python:

```
print(2*stats.t.cdf(-1.885, df=8))
0.0961
```

(notice the use of symmetry: $P(T > 1.885) = P(T < -1.885)$)

Conclusion: There is weak evidence against the null hypothesis (see Table 3.1); at $\alpha = 0.05$ we do not reject H_0.

The test statistic and p-value could also be calculated in Python (after having saved the raw data in the variable `x`) in the following way:

```
t_obs, p_val = stats.ttest_1samp(x, popmean=3000)
print(t_obs, p_val)

1.887 0.0957
```

b) At significance level $\alpha = 0.01$, what are the critical values for t_{obs}, and what is their interpretation?

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

Solution

The critical values are $\pm t_{0.995,\nu=8} = \pm 3.355$, as found using Python:

```
print(stats.t.ppf(0.995, df=8))
3.355
```

At $\alpha = 0.01$, H_0 is rejected only if t_{obs} falls outside the range $[-3.355, 3.355]$. Since the observed $t_{\text{obs}} = 1.885$ lies well within this interval, H_0 is not rejected.

c) At significance level $\alpha = 0.05$, what are the critical values for t_{obs}? And what is the interpretation of these critical values (compare also with the values found in the previous question)?

Use Python to compute quantiles and probabilities from relevant probability distribution(s).

|||| Solution

The critical values are $\pm t_{0.975,\nu=8} = \pm 2.306$, as found using Python:

```
print(stats.t.ppf(0.975, df=8))
2.306
```

Since $t_{\text{obs}} = 1.885$ is well within the interval $[-2.306, 2.306]$, H_0 is not rejected at $\alpha = 0.05$.

A smaller α corresponds to a larger *acceptance region* for t_{obs} (i.e., larger critical values): choosing $\alpha = 0.01$ demands stronger evidence against the null hypothesis H_0 than $\alpha = 0.05$ does.
Here "*stronger evidence against the null hypothesis*" corresponds to obtaining a sample average, $\bar{x}$, which is further away from the null hypothesis value μ_0 (which would lead to a larger t_{obs} - you can convince yourself of this by inspecting the formula for t_{obs}).

d) As mentioned in the beginning of the exercise the lengths of the concrete items are assumed reasonably normally distributed.
Is this assumption important for the results calculated above? Why?

|||| Solution

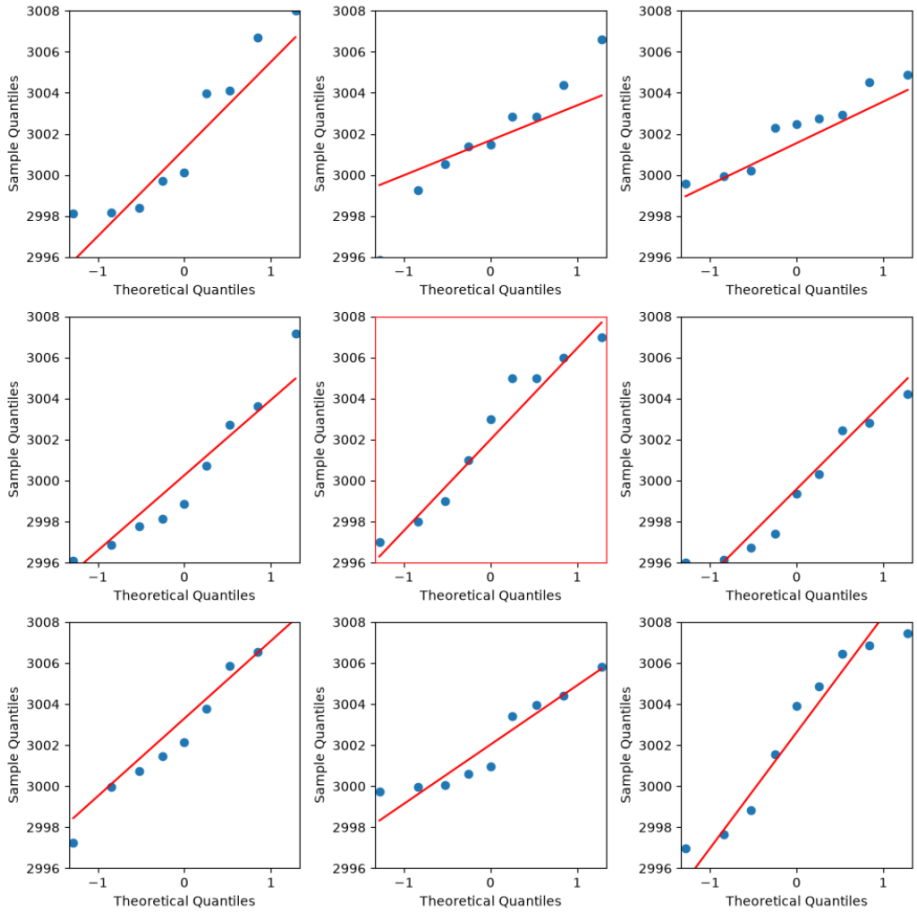
Yes, the normality assumption is important because the hypothesis test is based on the assumption that the estimator, namely the sample mean $\bar{x}$, follows a normal distribution. When the population of concrete item lengths is normally distributed, the sample mean is also normally distributed, and the test statistic follows a t-distribution. This is what allows us to calculate valid p-values and confidence intervals. Since the sample size is small ($n = 9 \leq 30$), this assumption is important for the validity of the results.

e) The construction company now investigates visually whether the data appears to be coming from a normal distribution (as assumed until now).

They produce a QQ-plot of the data and compare with QQ-plots of *8 simulated data samples* (all of size $n = 9$) sampled from normal distributions with $\mu = \bar{x}$ (mean equal to the sample mean) and $\sigma = s$ (standard deviation equal to the sample standard deviation), i.e., the simulated data-samples are all generated in the following way:

```
sim_data = stats.norm.rvs(size=9, loc=x.mean(), scale=x.
    std(ddof=1))
```

The QQ-plots looked like this (where the QQ-plot of the true data was plotted in the middle):



Does the plot give rise to concerns about the normal assumption?

Solution

No, the plot does not give rise to concerns about the normal assumption. The true data seems to follow the straight line equally well as the simulated data.

The code for constructing the plot is given here (if you want to try it yourself):

```
# Enter data into variable x:
x = np.array([3003,3005,2997,3006,2999,2998,3007,3005,3001])

# Generate 9 plots
fig, axs = plt.subplots(3, 3, figsize=(10, 10))

# Create 9 QQ plots of random normal samples (simulated
   normal data)
for i in [0,1,2]:
    for j in [0,1,2]:
        sim_data = stats.norm.rvs(size=9, loc=x.mean(),
           scale=x.std(ddof=1))
        sm.qqplot(sim_data, line="q", ax=axs[i,j])
        axs[i,j].set_ylim([2996, 3008])

# Replace middle plot with real data
axs[1,1].clear()
sm.qqplot(x, line="q", ax=axs[1,1])
axs[1,1].set_ylim([2996, 3008])

# Highlight middle plot with red spines
plt.setp(axs[1,1].spines.values(), color="red")

# Generate the plot
plt.tight_layout()
plt.show()
```